PRINCIPLES COURSE • APRIL 22, 2026

ESi

Vehicle-Track Measurement Technologies

DALLAS, TX

Summary

Overview of Measurement Systems

Detailed Evaluation of Technology Platforms

Simulation Tools for Derailment Investigation

Overview of Measurement Systems

There are six basic categories of measurement system.

Categories of Measurement Systems

1) Mounted on <u>Vehicle</u> to measure the <u>Vehicle.</u>	

Image from Amsted Digital Solutions

Image from HUM Industrial

Categories of Measurement Systems

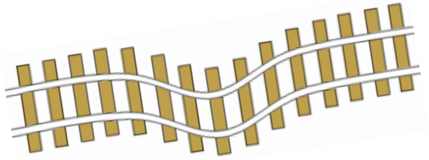
2) Mounted on Track to measure the Track.	

Image from FTS Tools

Ref: https://www.voestalpine.com/railway-systems/en/products/zentrak-switch-condition-monitoring/

Categories of Measurement Systems

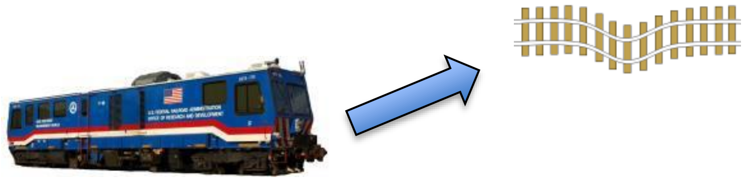
3) Mounted on <u>Vehicle</u> to measure the <u>Track.</u> (Manned, Unmanned, and Autonomous)	

Image from Holland

Categories of Measurement Systems

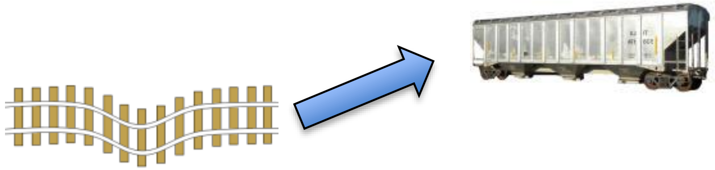
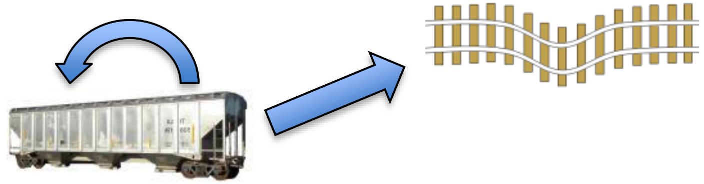
4) Mounted on Track to measure the Vehicle.	

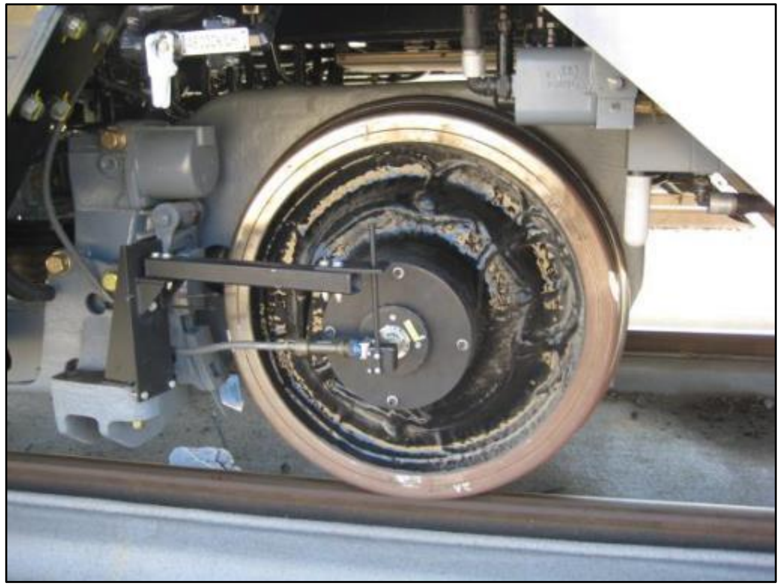
Image from L.B. Foster

Image from KLD Labs, Inc.

Categories of Measurement Systems

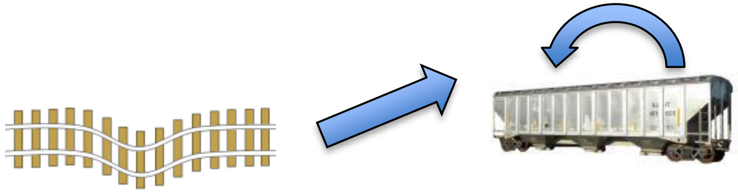
5) Mounted on Vehicle to measure the Vehicle & Track.

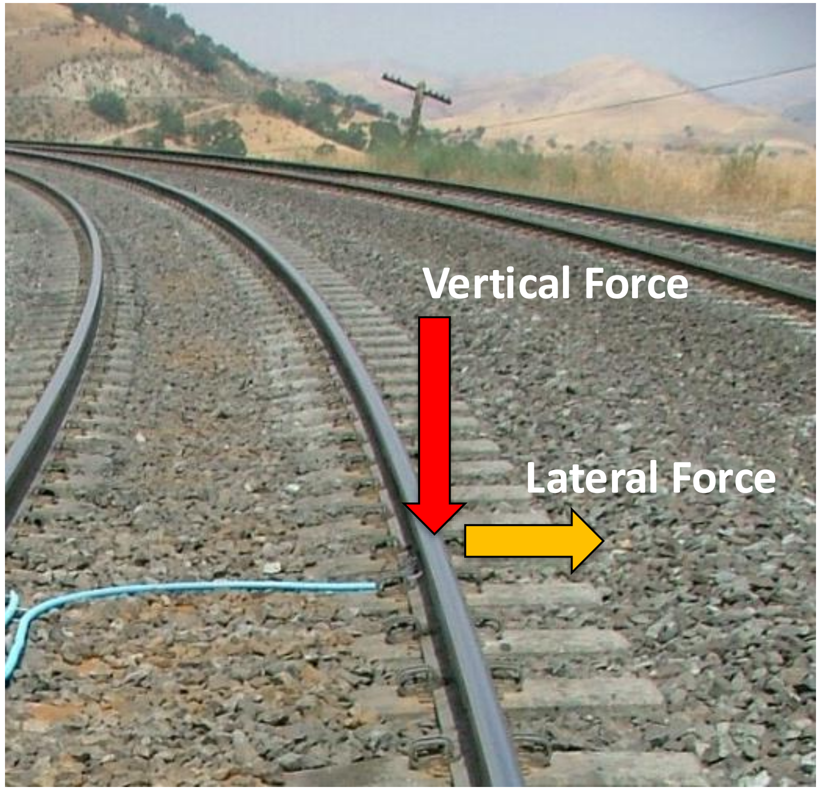




Images from ENSCO Rail, Inc.

Categories of Measurement Systems

6) Mounted on Track to measure the Vehicle & Track.	



Truck Performance Detectors

Also known as "L/V Detector"

Common Uses:
Vehicle condition monitoring
Superelevation assessments
New Vehicle Fleet Assessments

Track Measurement		Vehicle Measurement
Track Geometry Measurement	⟷	Truck Condition Monitor (TBOGI)
Rail Profile Measurement System	⟷	Wheel Profile Detector
V/TI Monitor Axle Impact	⟷	Wheel Impact Load Detector
2D & 3D Machine Vision	⟷	2D & 3D Machine Vision

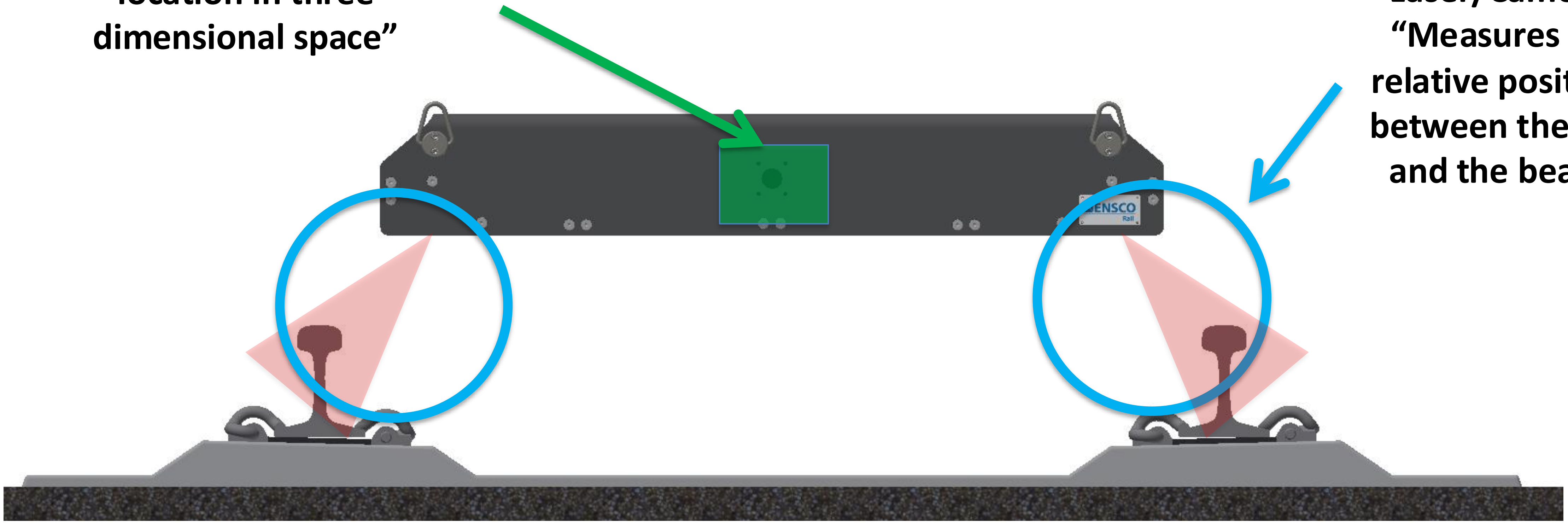
Other Systems Important for Derailment Investigation

Geometry Measurement

Track Geometry Measurement System

Inertial Package
“Measures the beam location in three dimensional space”

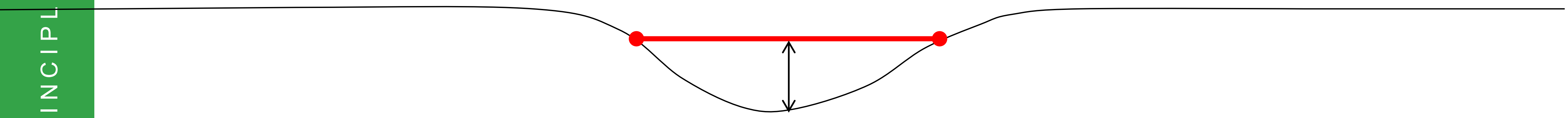
Laser/Cameras
“Measures the relative positions between the rails and the beam”



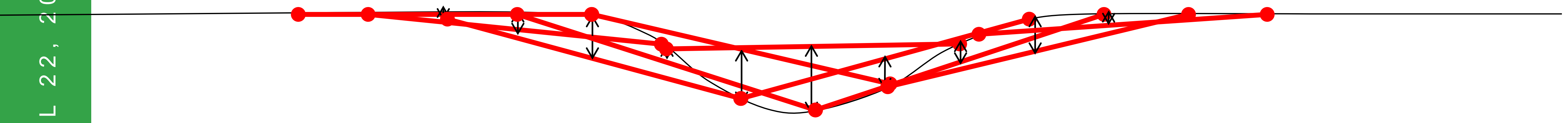
Space Curve vs. MCO

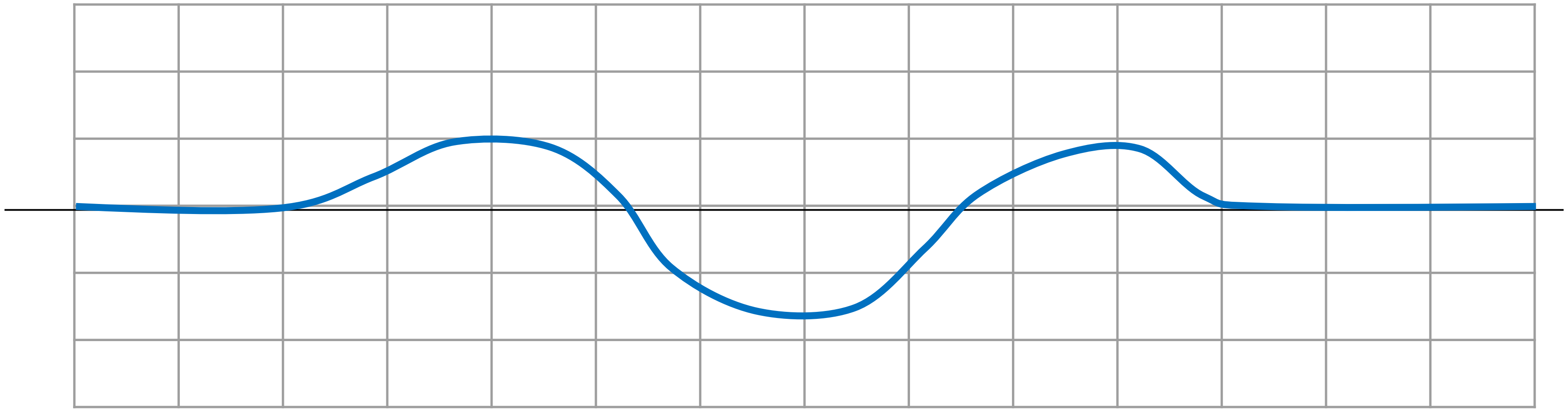
Space Curve is intended to represent the true geometry of Profile or Alignment conditions.

Mid-Chord Offset is used to quantify Profile or Alignment conditions.



Space Curve vs. MCO





Common Profile and Alignment Chord Lengths:

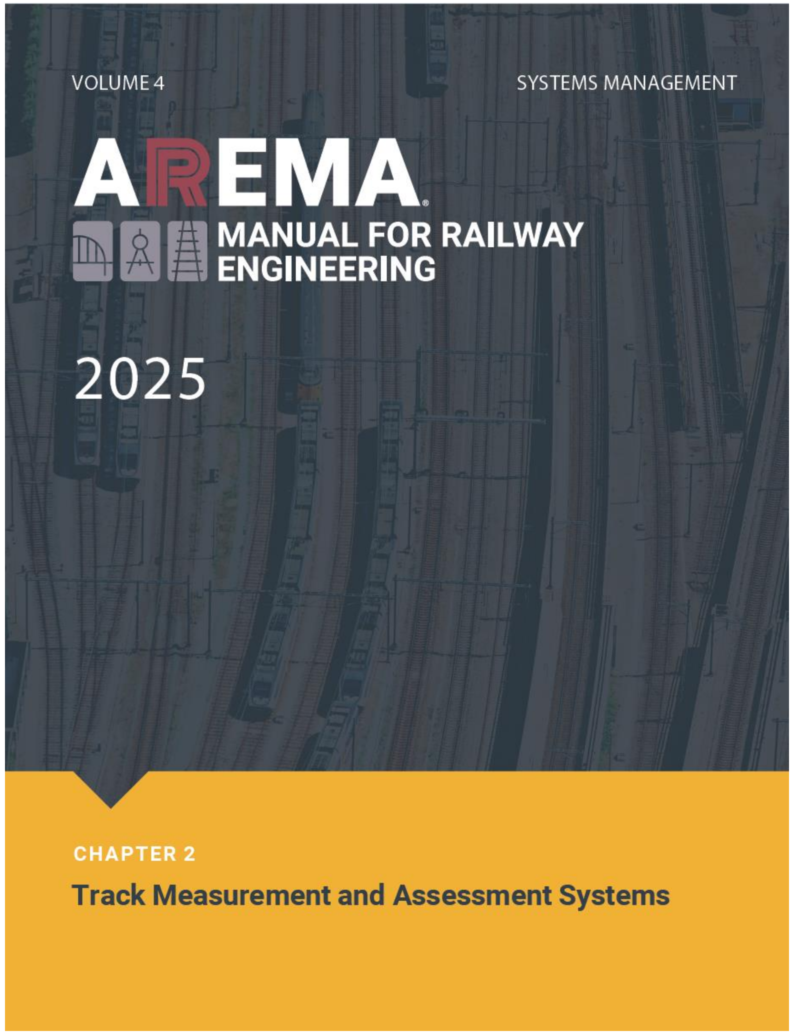
10 ft

31 ft

62 ft

124 ft

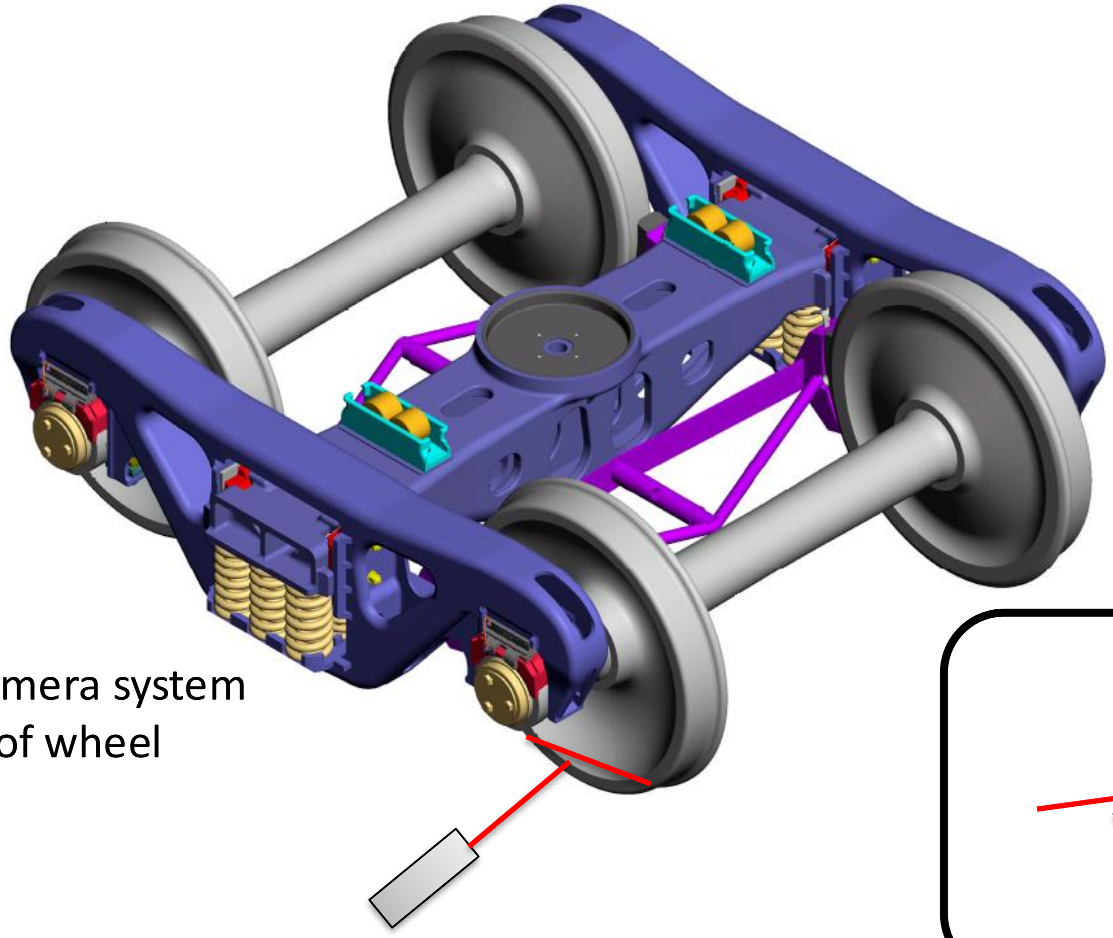
Space Curve vs. MCO



AREMA Chapter 2 now has a great definition of Space Curve and MCO (Part 3 Track Geometry Measurement)

Also New:
SECTION 9.3 RECOMMENDED PRACTICE FOR USING TRACK GEOMETRY MEASUREMENT SYSTEM DATA FOR VEHICLE/TRACK INTERACTION MODELING

Truck Condition Monitor (T-BOGI)



Wayside Laser/Camera system measures profile of wheel plate.

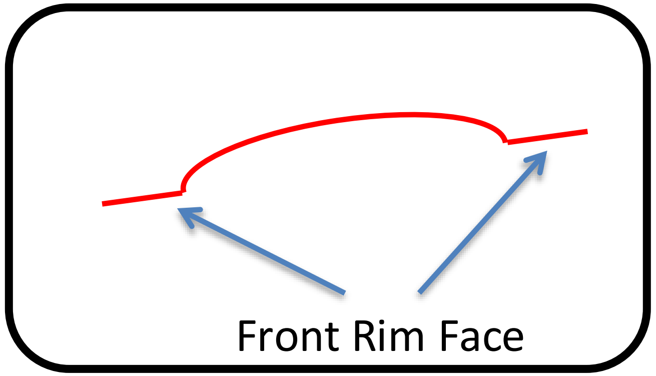
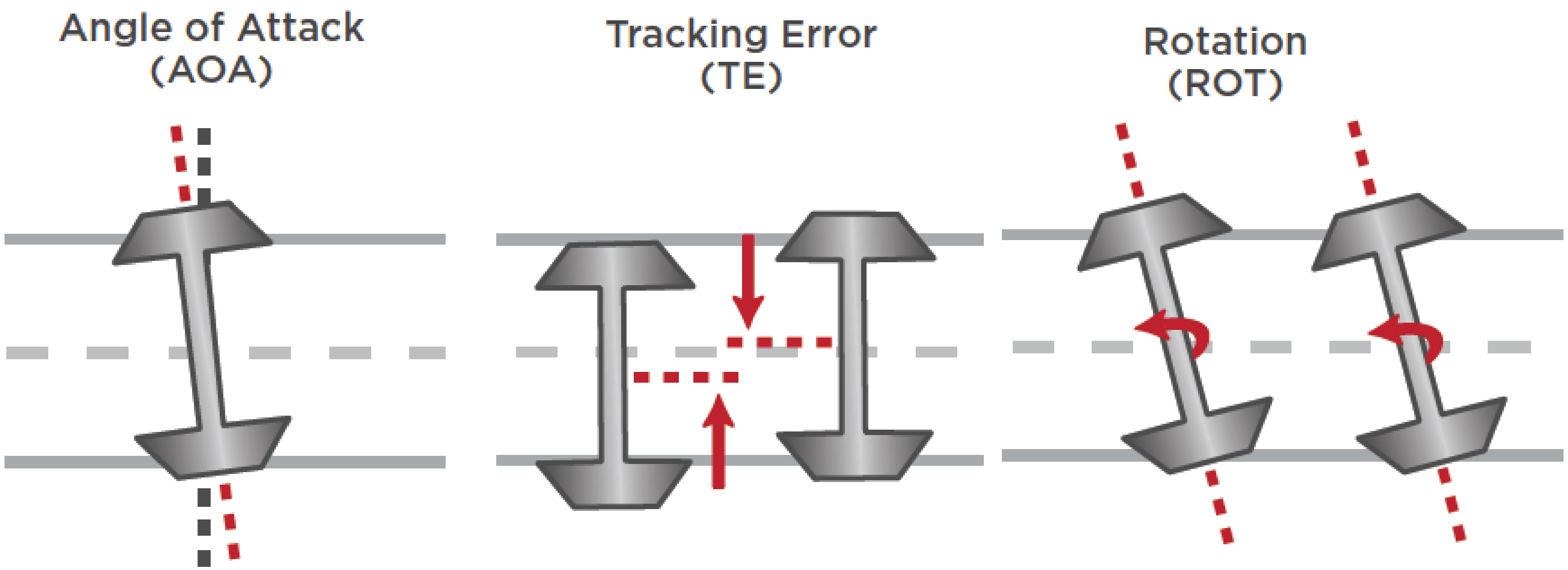


Image from Standard Car and Truck Maintenance Manual
http://www.sctco.com/pdf/Section1.pdf

How does a truck condition monitor work?



Images from Wayside Inspection Systems
http://wid.ca/sites/default/files/brochures/TBOGI/WID_TBOGI_Brochure_US.pdf

Profile Measurement

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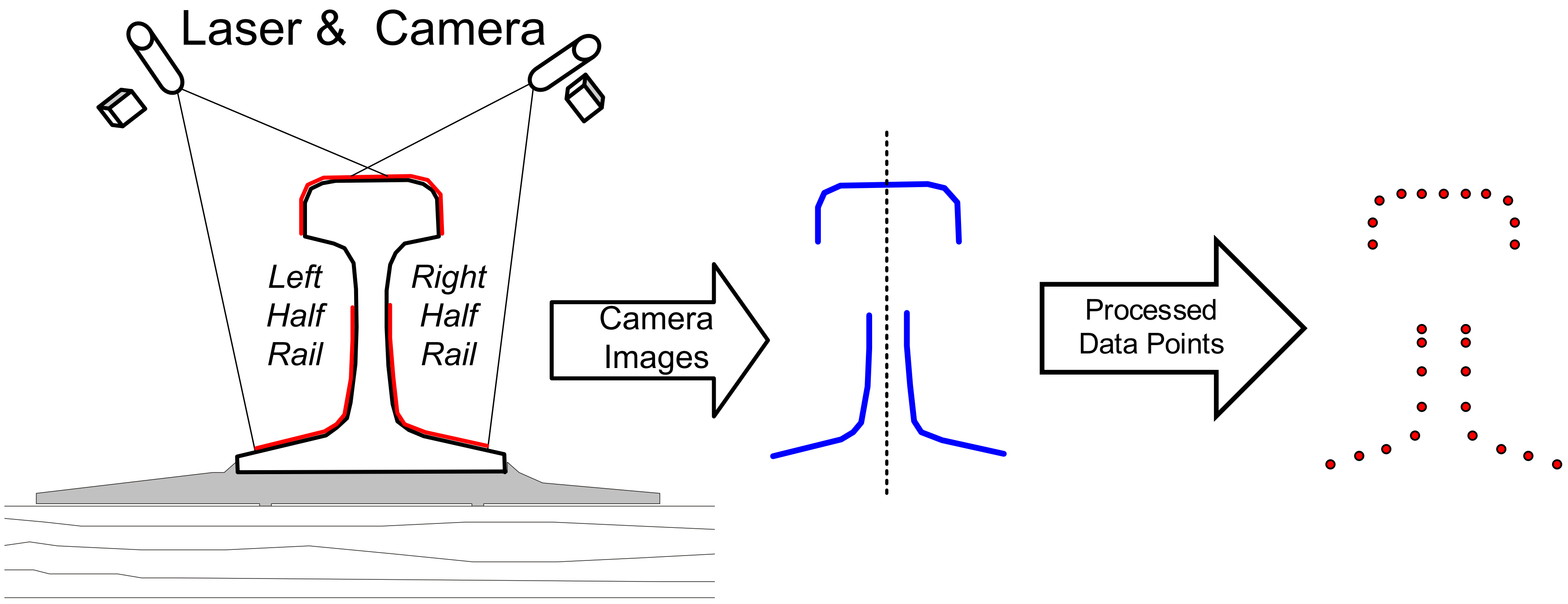
Laser Triangulation Measurement



Ref 1

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Laser Triangulation Measurement

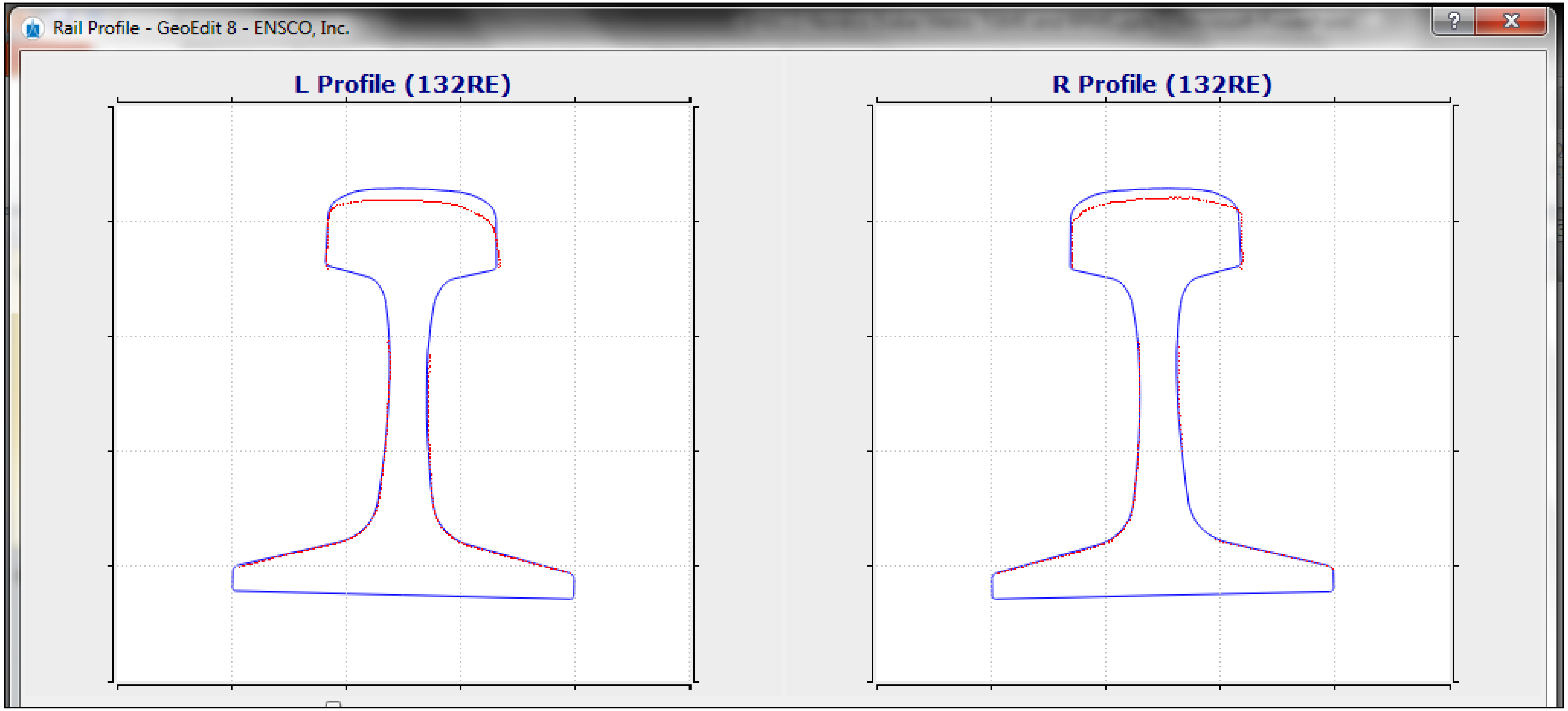
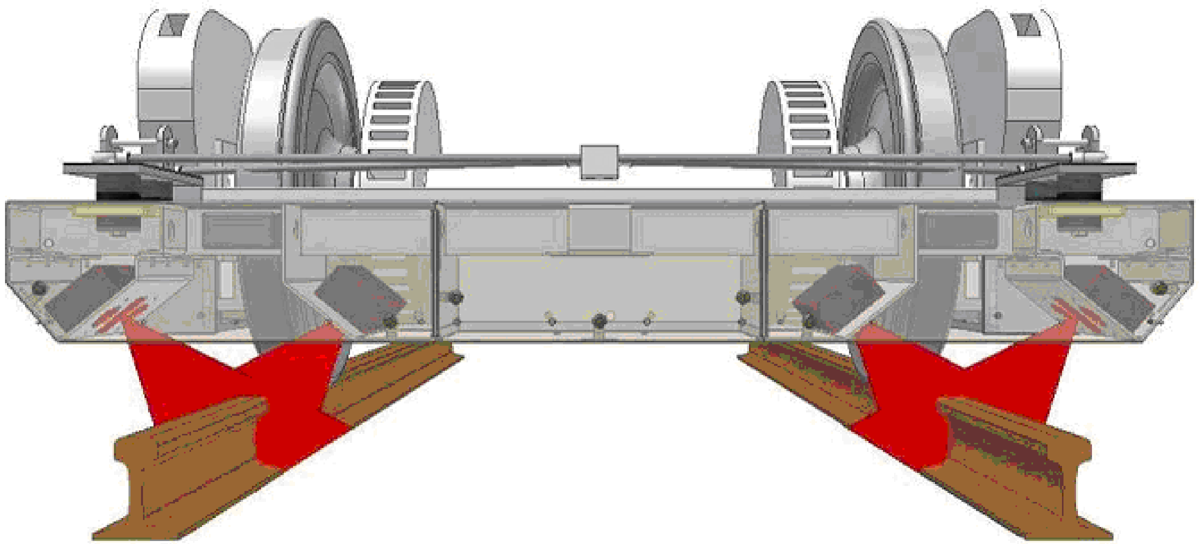


Image from ENSCO Rail, Inc.

Rail Profile Measurement System

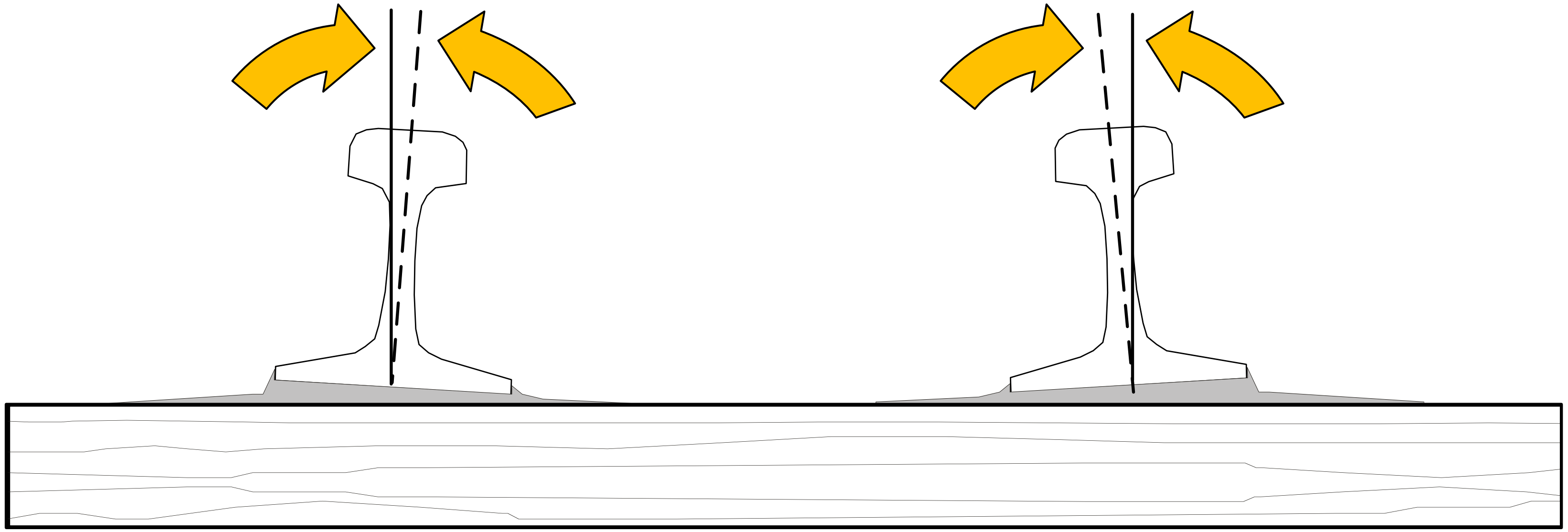


Images from ENSCO Rail, Inc.

Absolute Measurements:

(Doesn't Require Template)

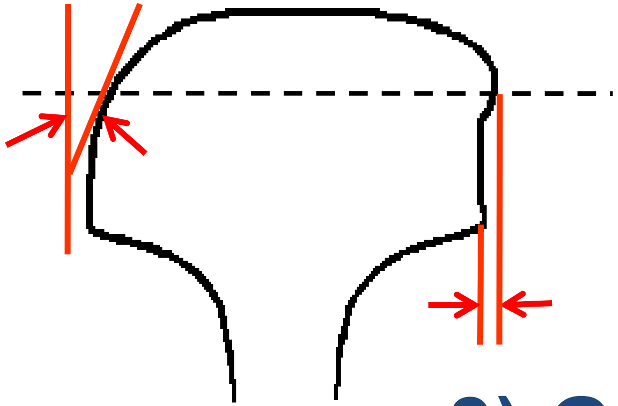
1) Cant (Deg)



Absolute Measurements:

(Doesn't Require Template)

2) Gage Face Angle (Deg)

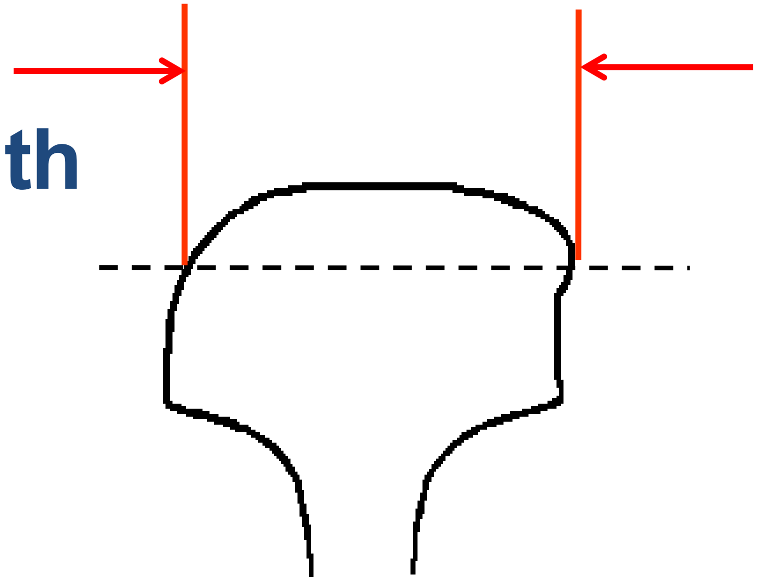


3) Gage/Field Side Lip (In, mm)

Absolute Measurements:

(Doesn't Require Template)

4) Head Width (in, mm)

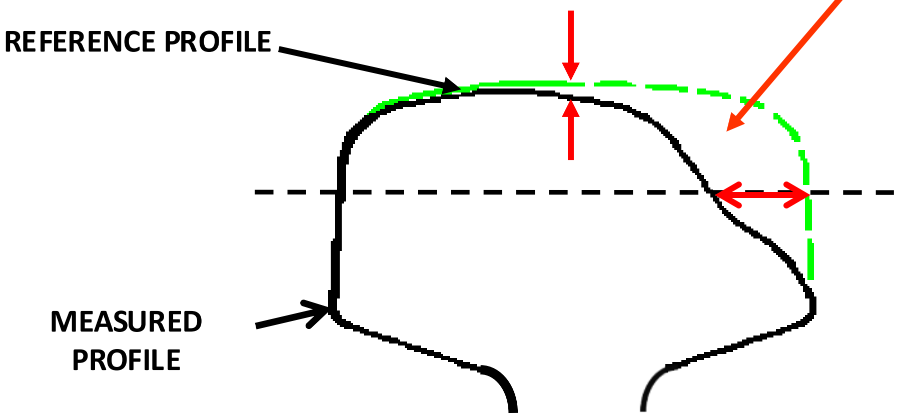


Relative Measurements:

(Does Require Template)

5) Vertical Head Wear (in, mm)

6) Head Loss (Sq inch, sq mm & %)

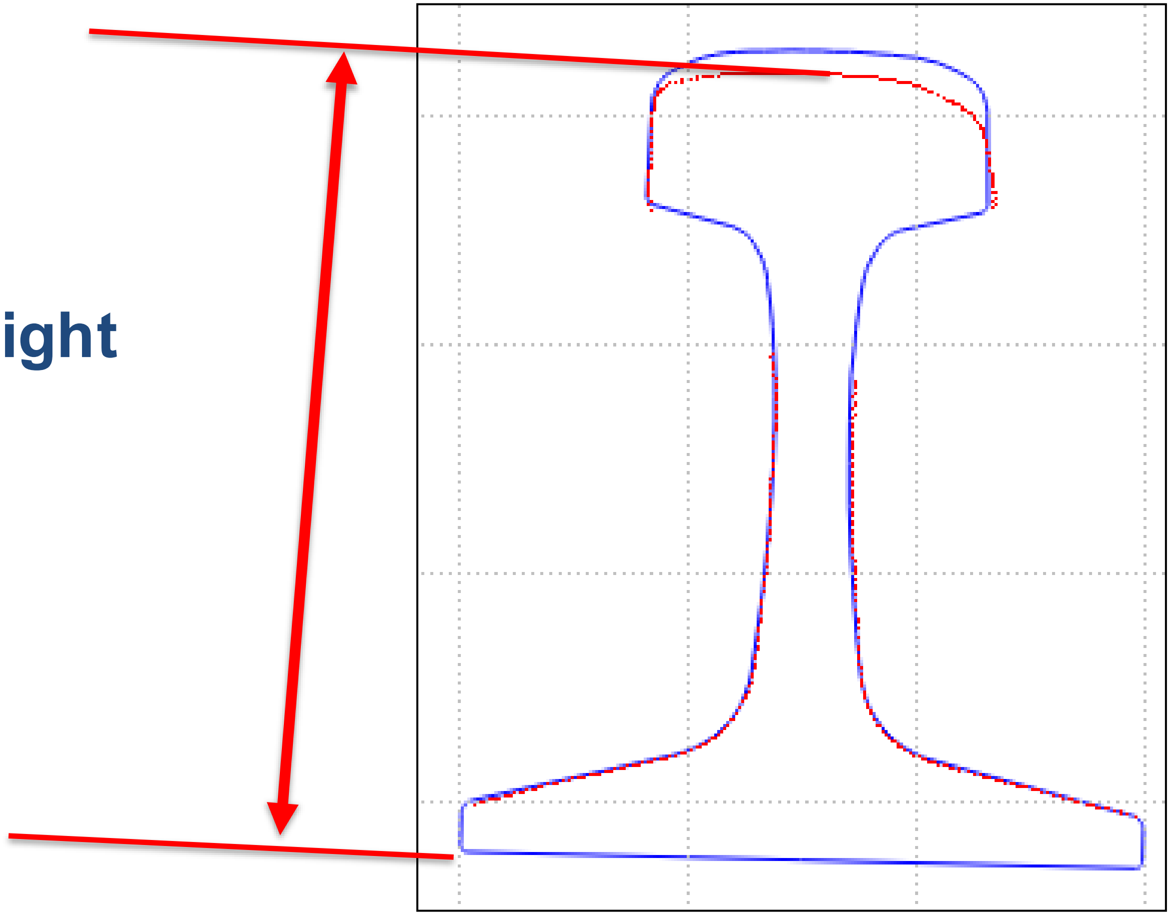


7) Gage/Field Wear (in, mm)

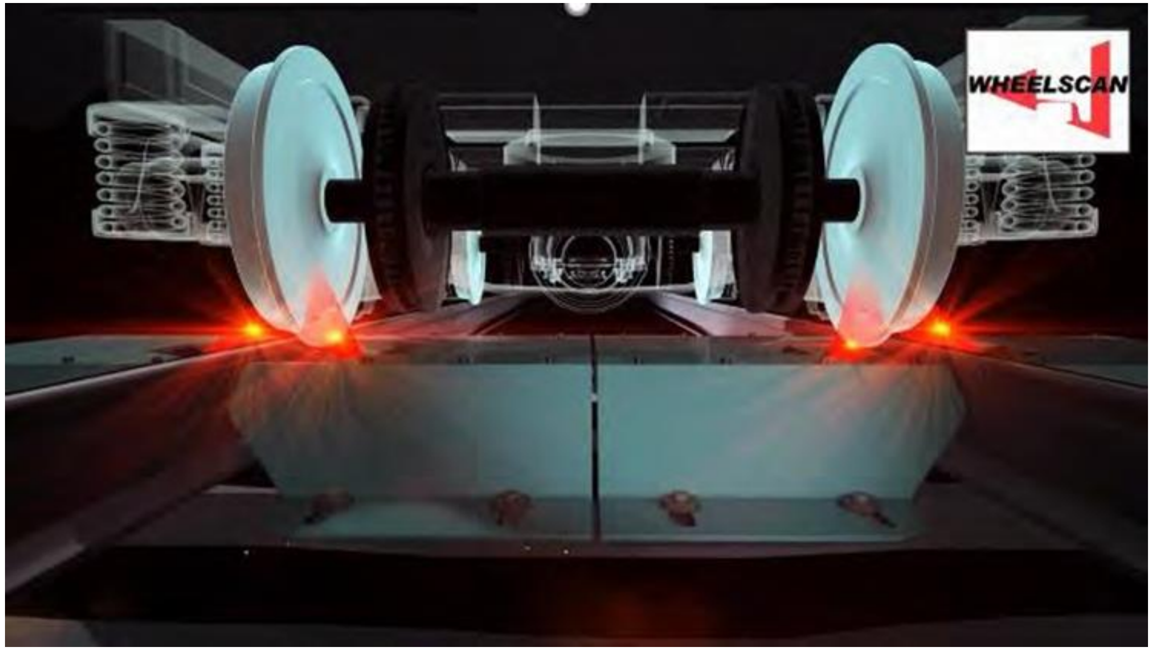
Relative Measurements:

(Does Require Template)

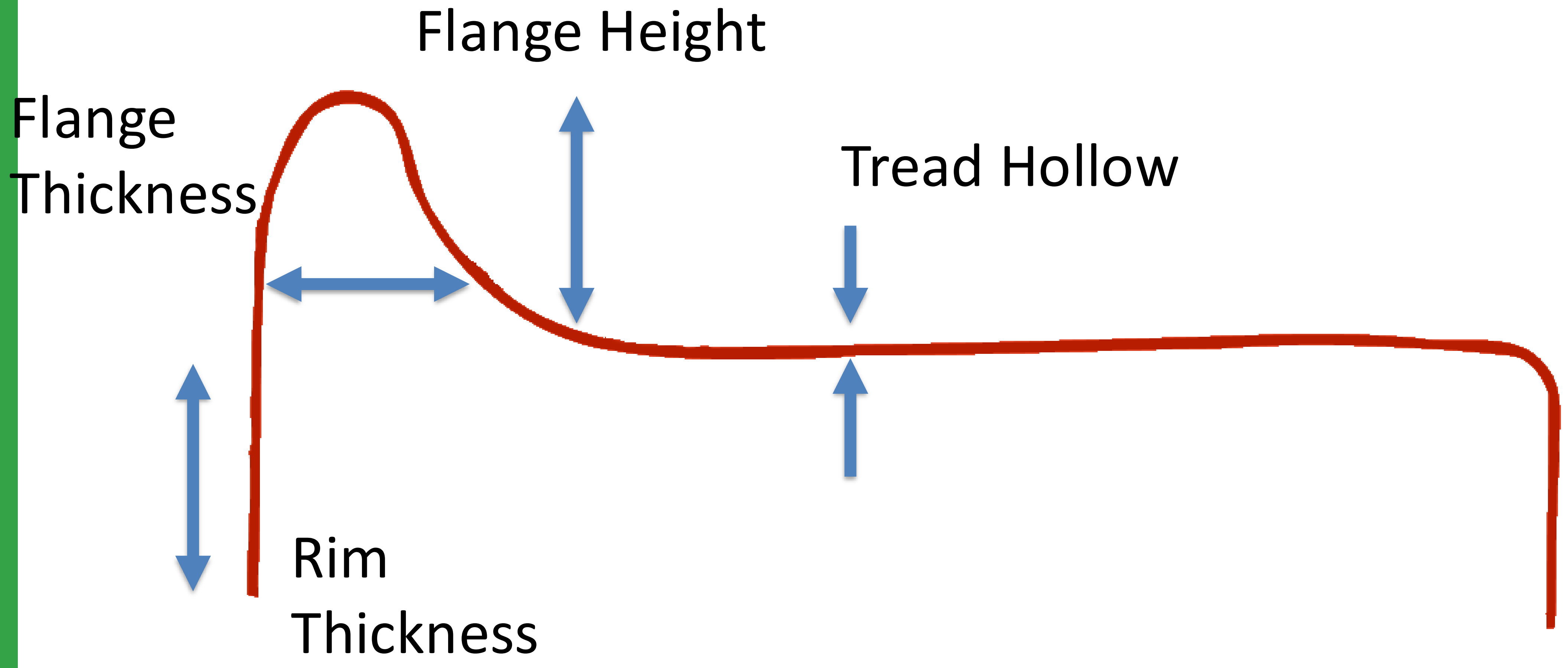
8) Total Height (in, mm)



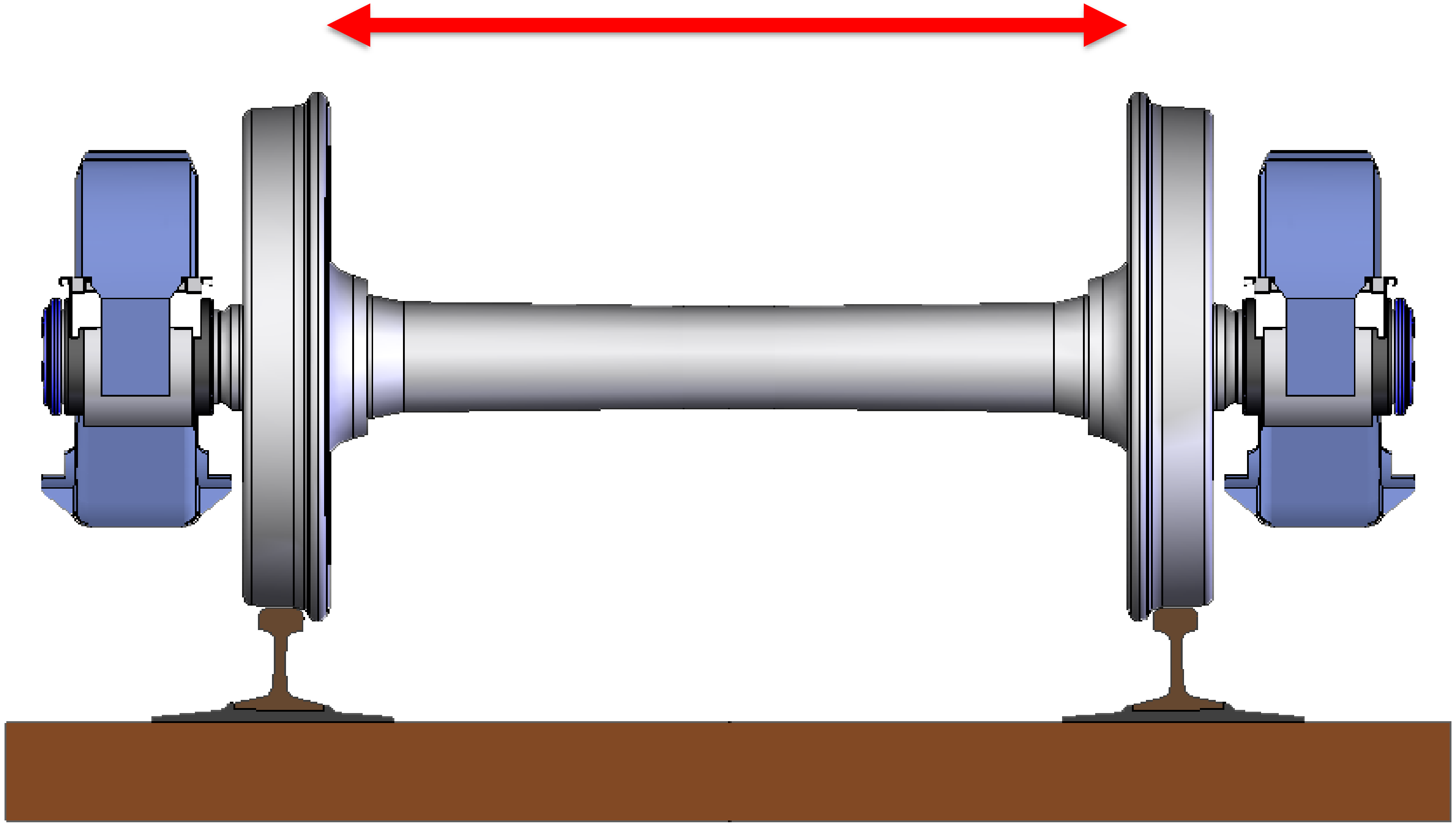
Wheel Profile Detector



Images from KLD Labs, Inc.


Flange Height
Flange
Thickness
Tread Hollow
Rim
Thickness

Back-to-Back



Impact Measurement

Wheel Impact Load Detector (WILD)

Image from L.B. Foster
https://www.youtube.com/watch?v=7CJycCggHgw

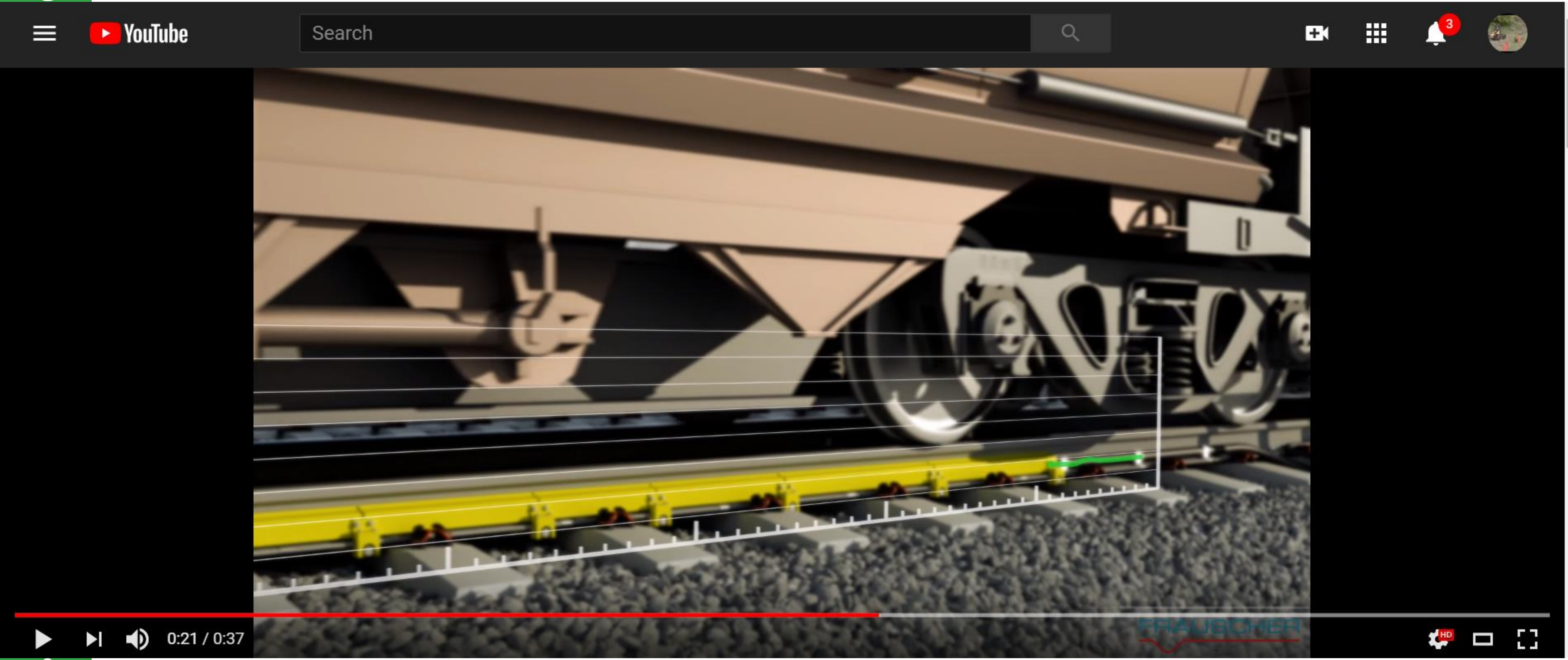
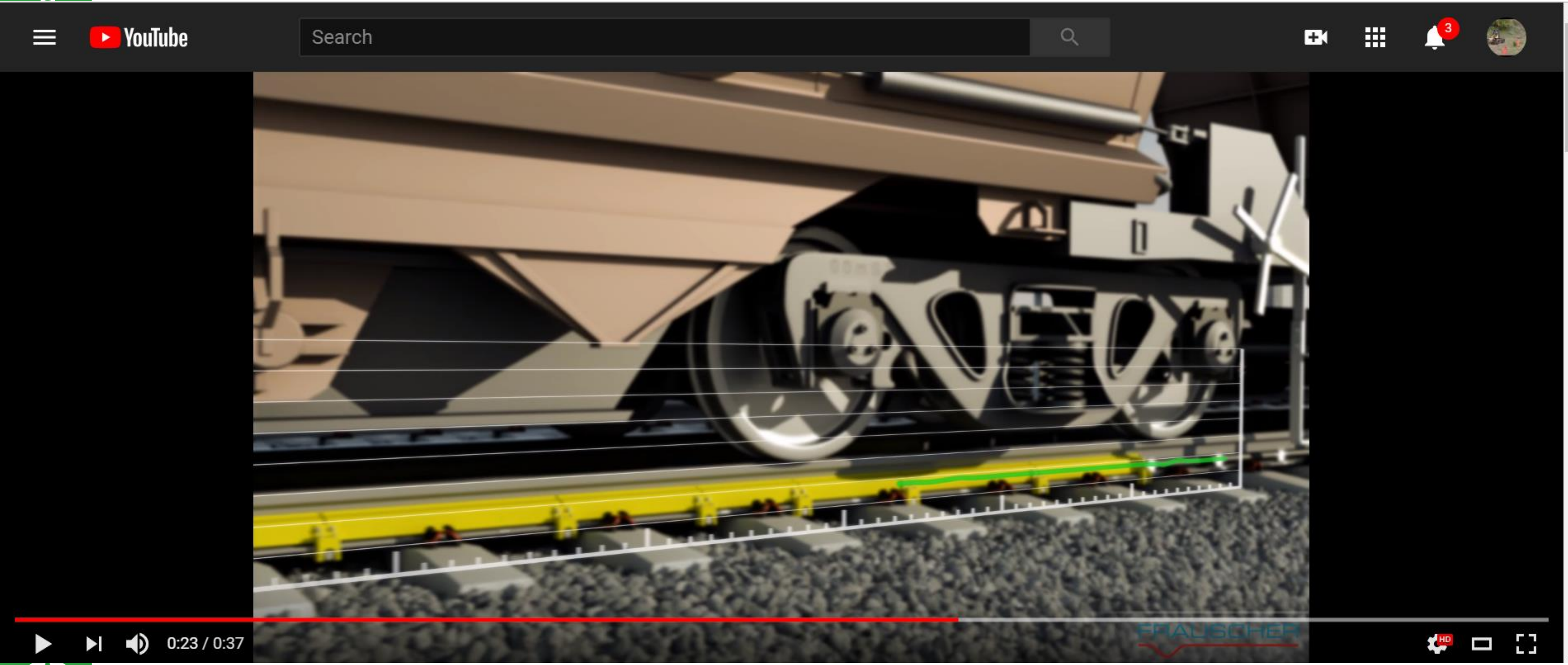


Image from Frauscher
https://www.youtube.com/watch?v=gTfU4tGZzgo



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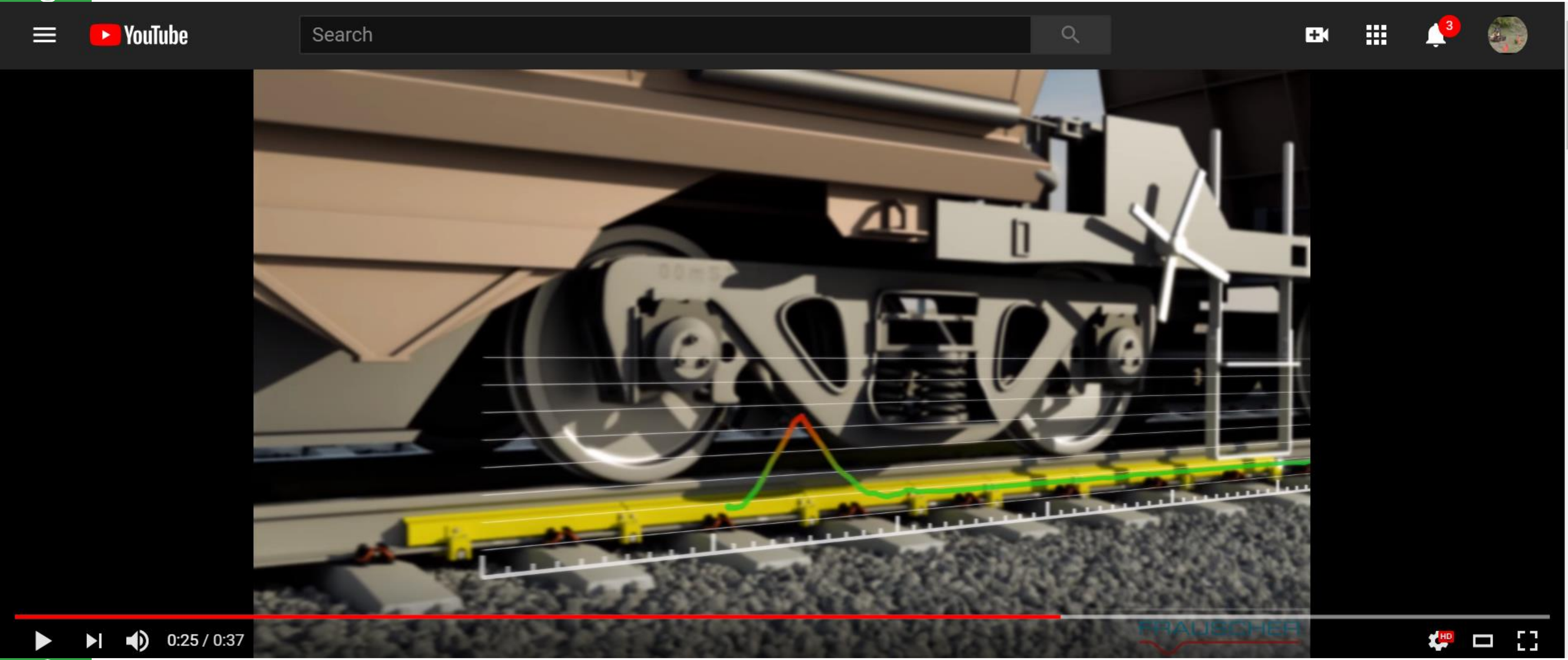
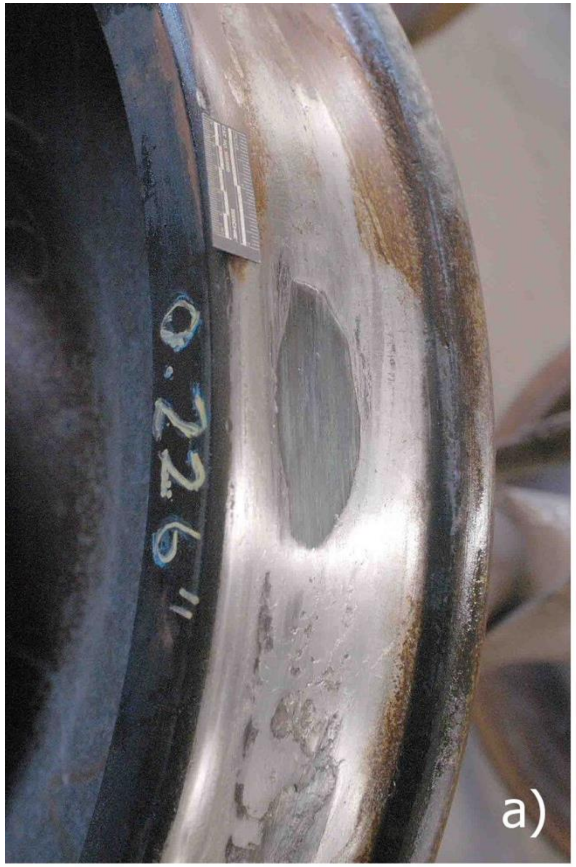
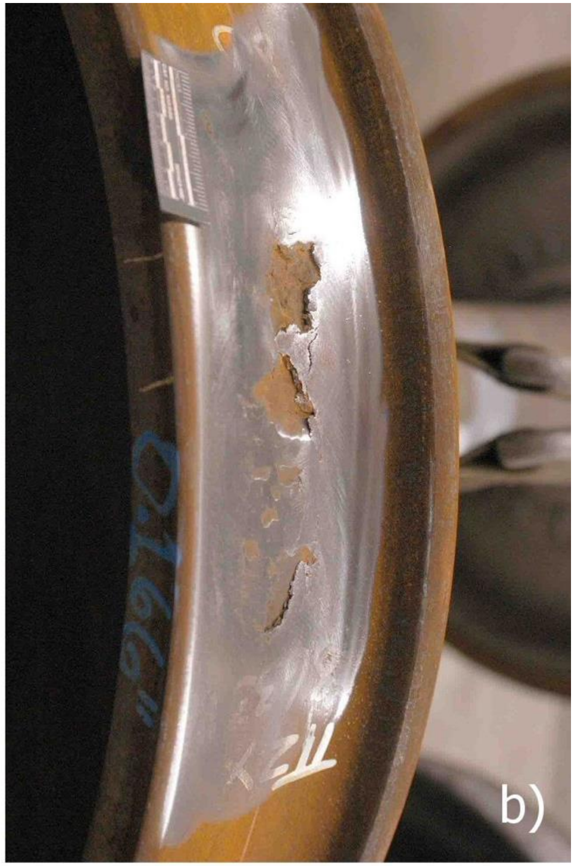


Image from Frauscher
https://www.youtube.com/watch?v=gTfU4tGZzgo

Example WILD Defects

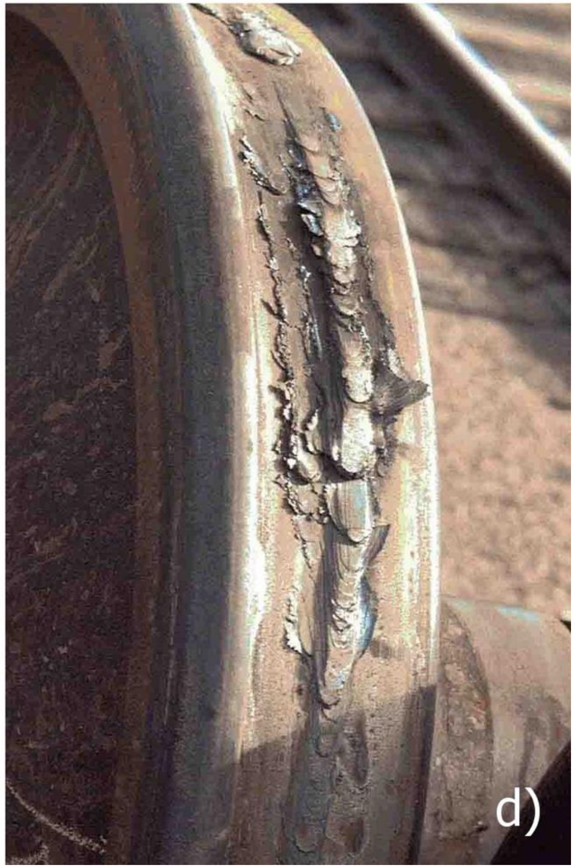


Slid Flat



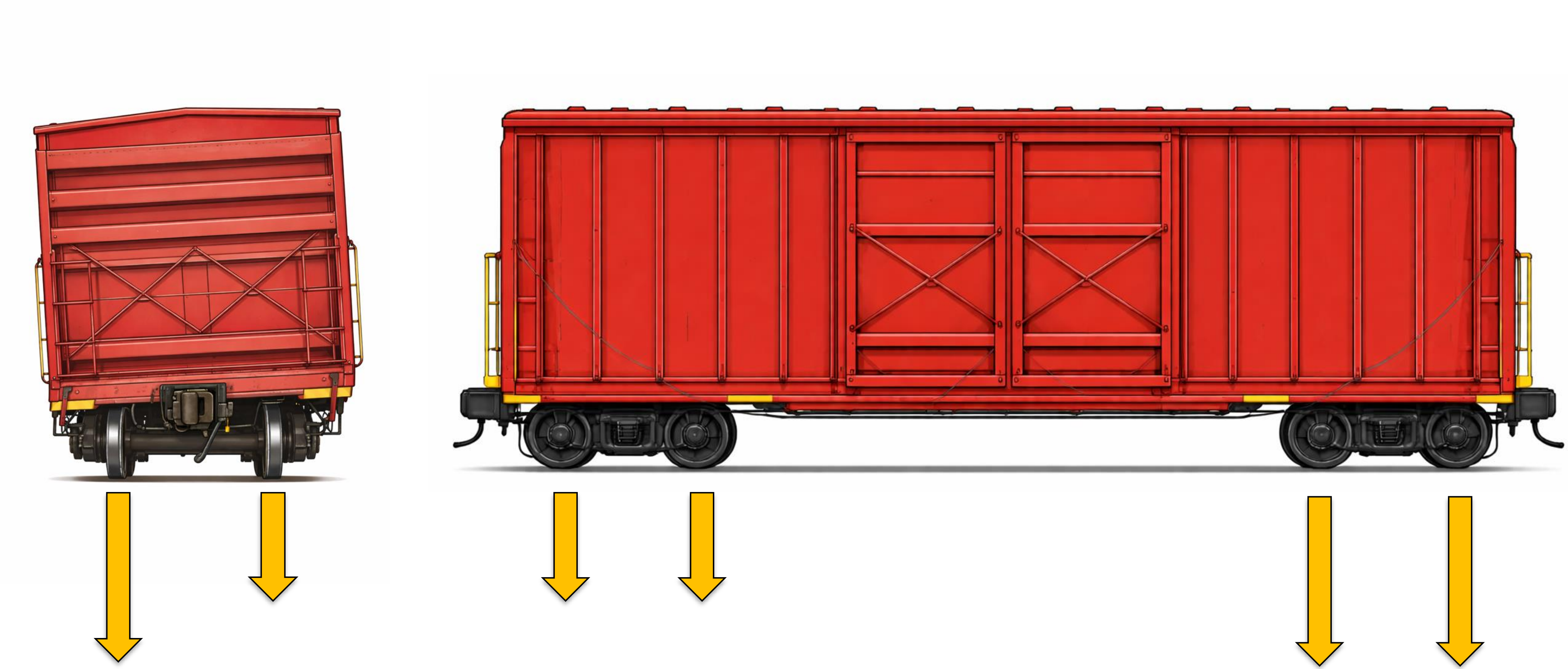
Shelling & Spalling

Shattered Rim



Built Up Tread

Shifted Loads Measured by WILD



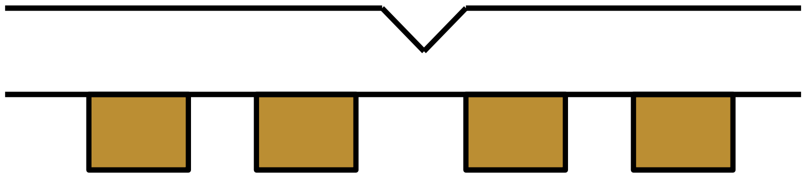
V/TI Monitor – Impact Measurement

Image from ENSCO Rail, Inc.

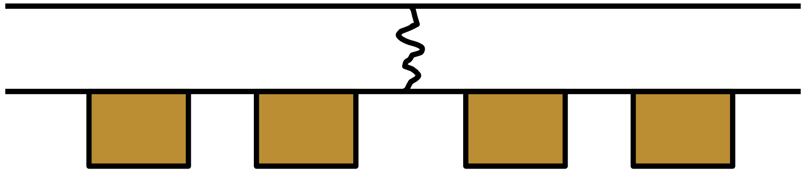
Example V/TI Monitor Axle Sensor

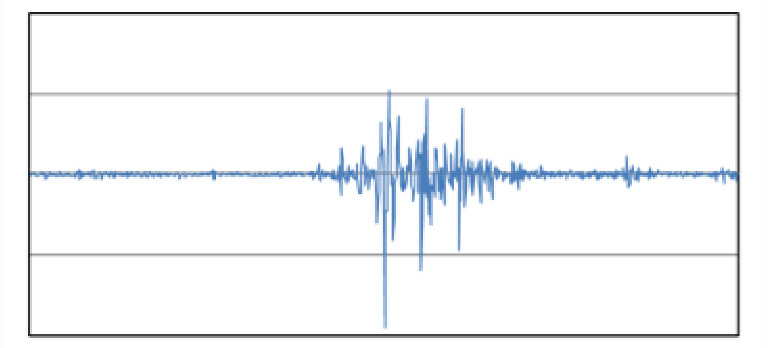
V/TI Monitor – Impact Measurement

Dent



Break





Vertical acceleration measured at axle box

Wheel/Rail Impact acceleration measured

Peak acceleration is used to calculate peak load.

Example V/TI Monitor Defects

Battered Joint

Cracked/Broken Joint Bar

Cracked/Broken Frog

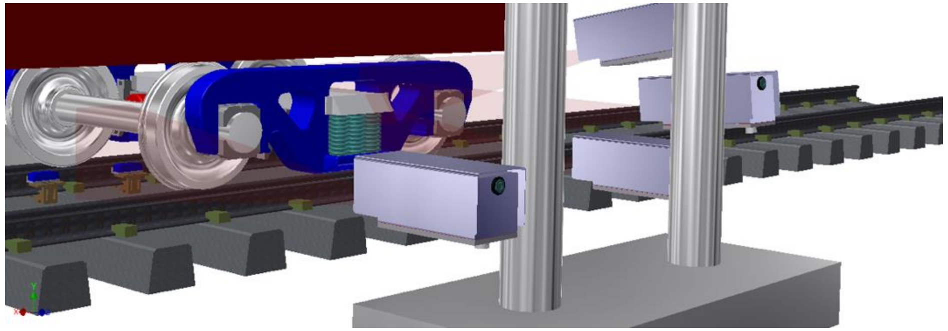
Broken Rail

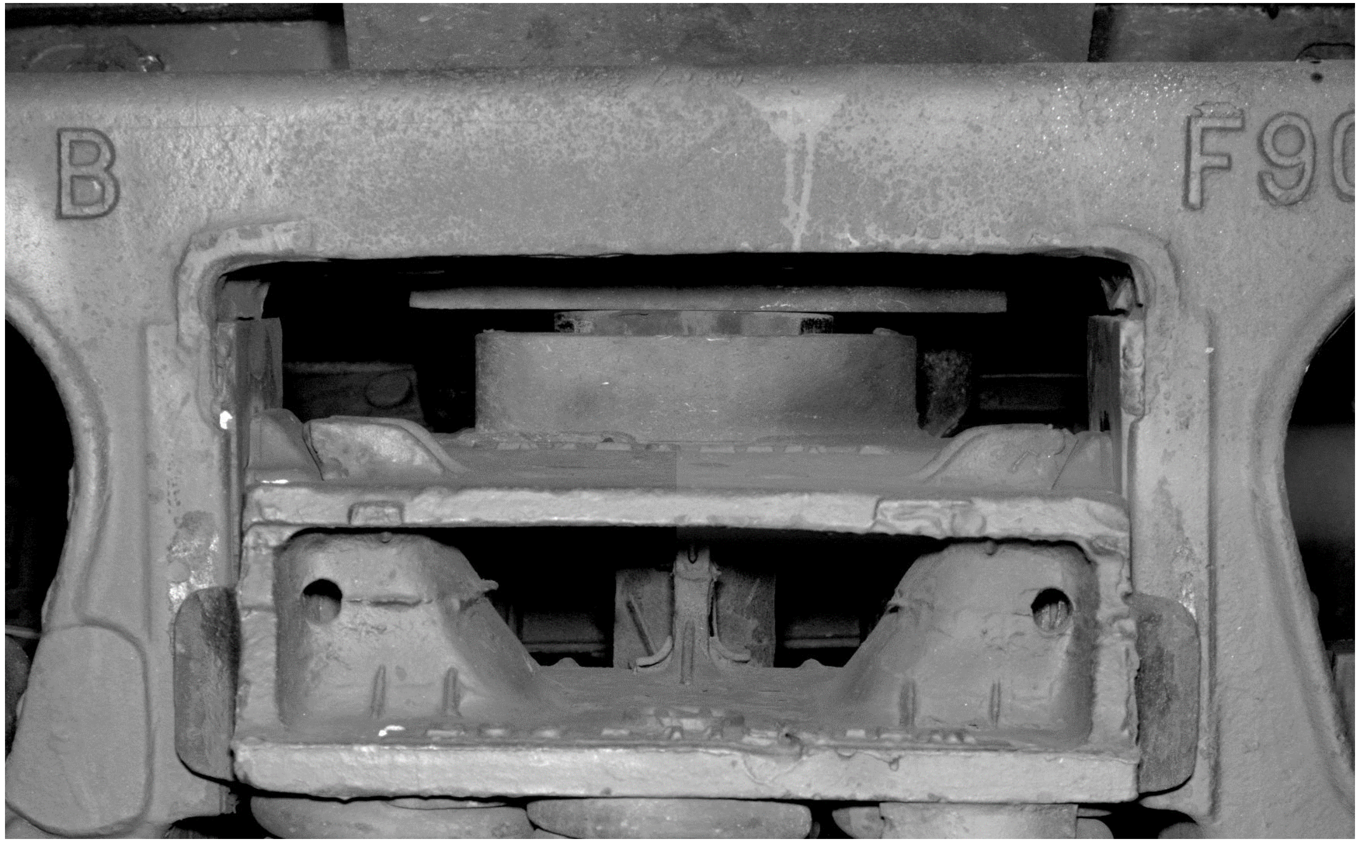
2D and 3D Machine Vision

Friction Wedge Rise Measurement

2D Area or Line Scan Cameras

Identifies friction wedge rise





Images from KLD Labs, Inc.

Rail Surface RCF

2D Line Scan Cameras

Identifies RCF or other Rail Surface Damage

Curvature (deg)

Crosslevel (in)

Gage (in)

L Cant (deg)

R Cant (deg)

L FS Crack Density

R FS Crack Density

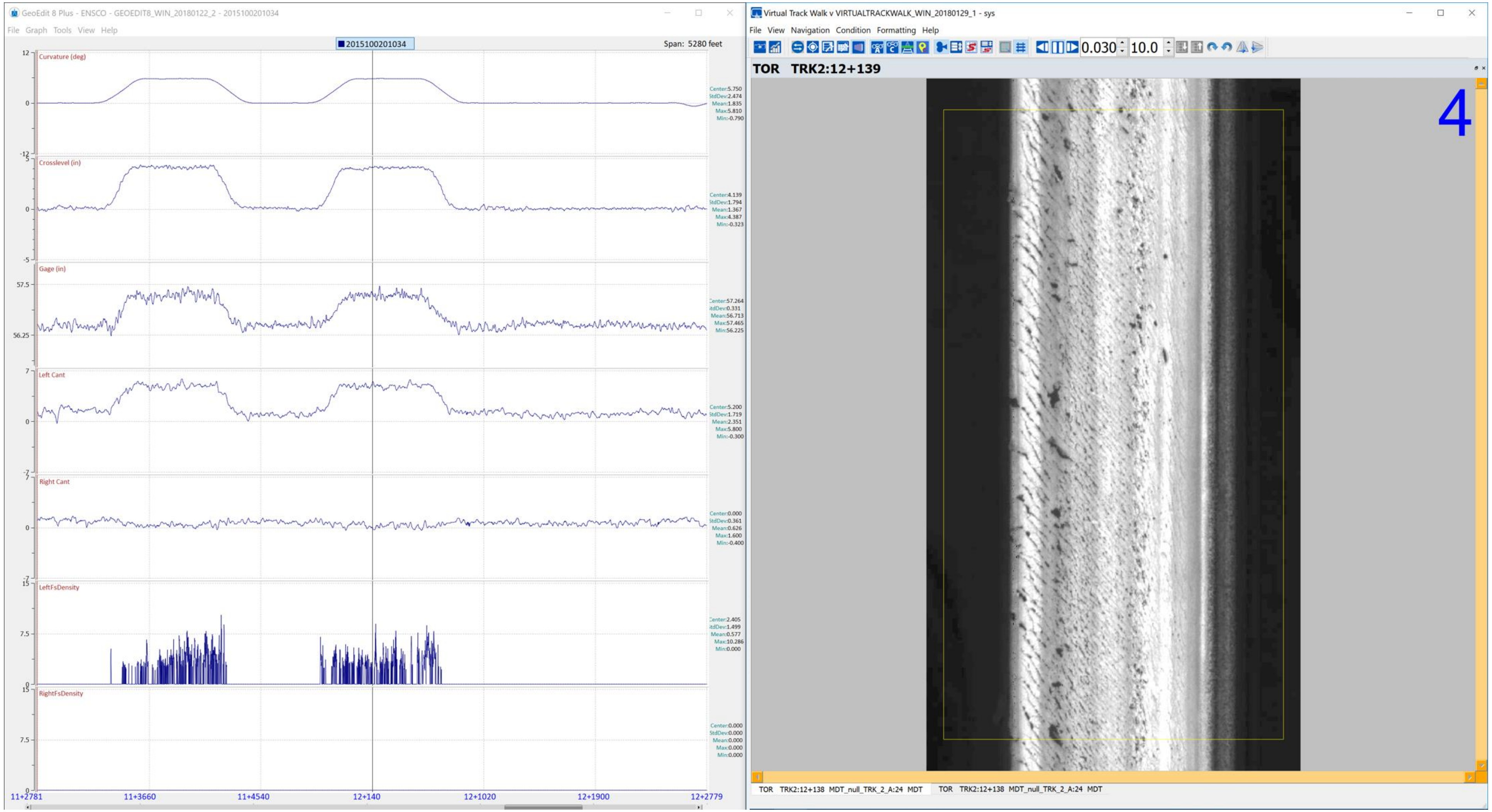


Image from ENSCO Rail, Inc.

3D Track Scans

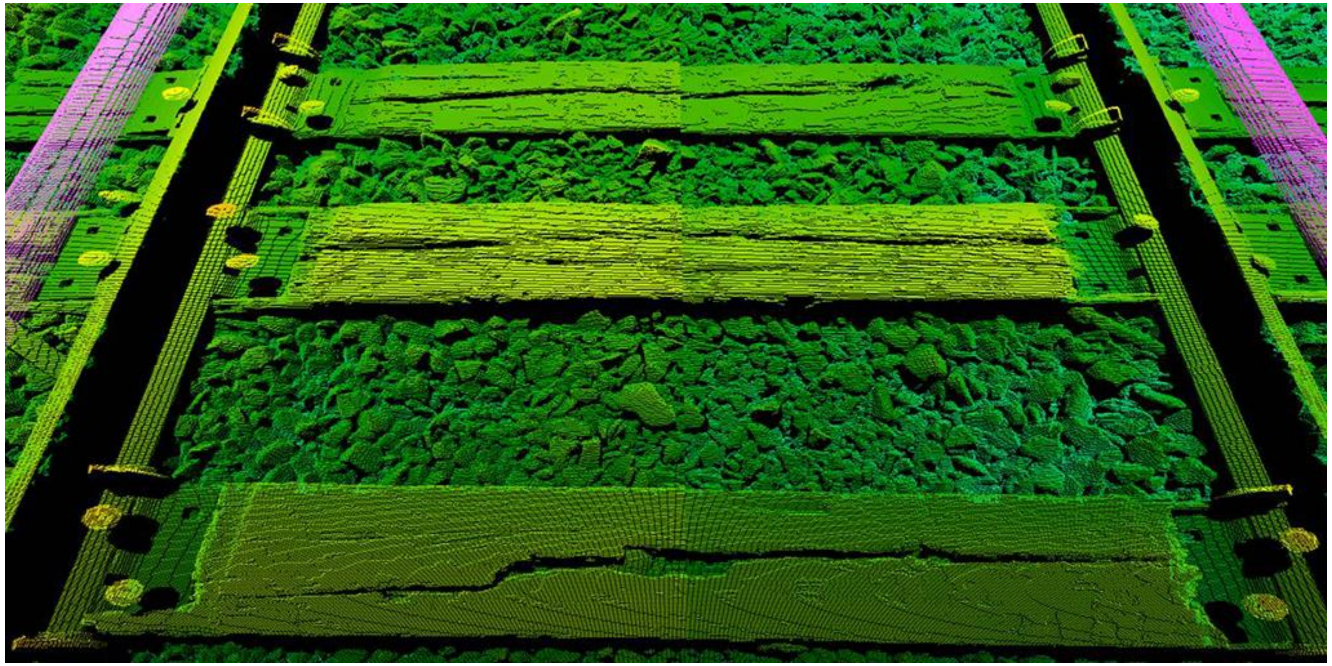


Image from Loram

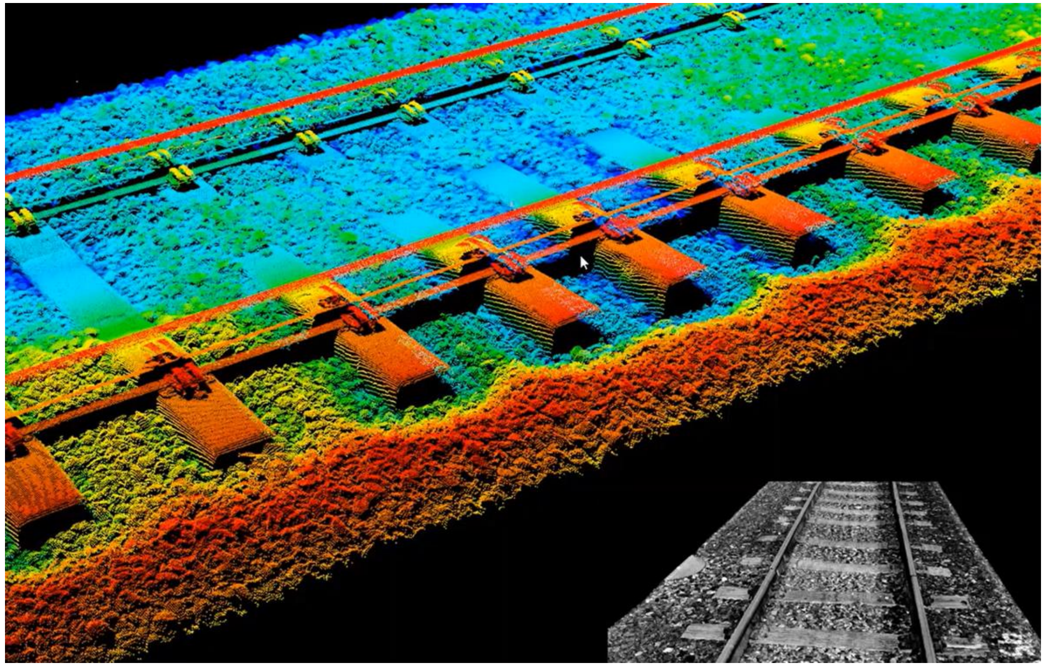


Image from Railmetrics

3D LiDAR Point Clouds

Image from Tetra Tech

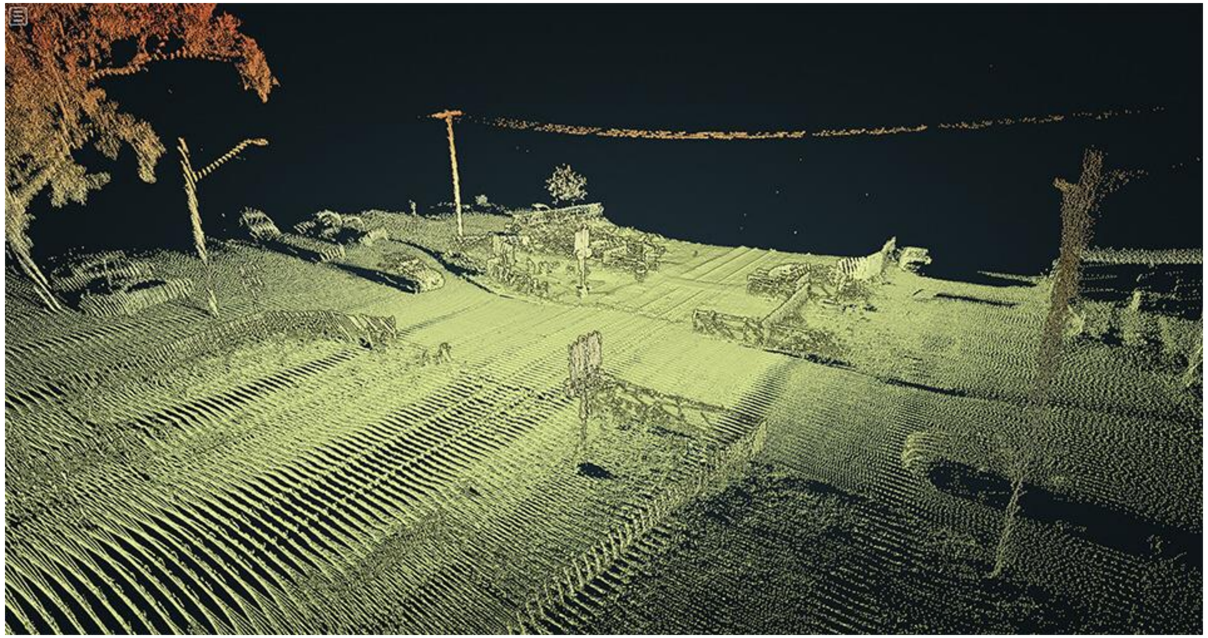


Image from Cordel

Other Systems Important for Derailment Investigation

Ultrasonic Rail Flaw & Eddy Current

Associated to **broken rail derailments.**

Rail Flaw survey records are checked as a default, if even to rule out as a derailment factor.

Hot Bearing Detectors (HBD) & Acoustic Bearing Detectors (ABS)

Associated to **bearing burn off derailments**

HBD & ABD records are checked as a default, if even to rule out as a derailment factor.

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Handheld Wheel and Rail Profile Measurement

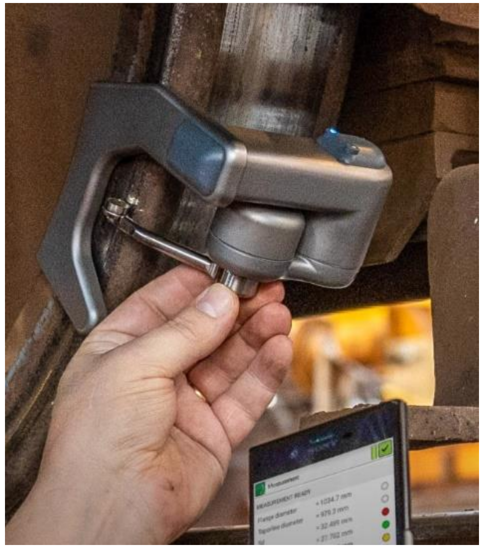


Image from Greenwood Engineering

Image from Hexagon

Derailment Simulations

Simulation Types

Track-Train Dynamics Simulations (TOES or TEDS)

Vehicle/Track Interaction Simulations (NUCARS or VAMPIRE)

Track/Train Dynamics Simulations

Important Items for Simulation:

- Get the MP and GPS coordinates of where the head-end locomotive stopped.
- Measure the wheel diameter of the encoder wheel of the locomotive.
- Download the event recorder information of all locomotives in consist.
- Determine if any of the locomotive has cut out traction motors
- Get the train consist information including:
  - Locomotive Positions
  - Car Positions with Initials and Numbers
  - Car Weights (actual weight, not rated or tare)
  - Car Lengths

Vehicle/Track Interaction Simulations

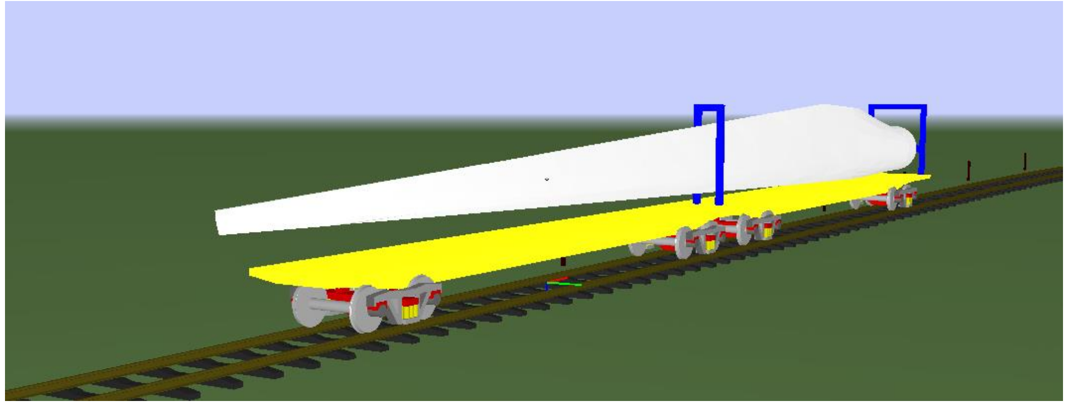
Simulates rail vehicles travelling over track features.

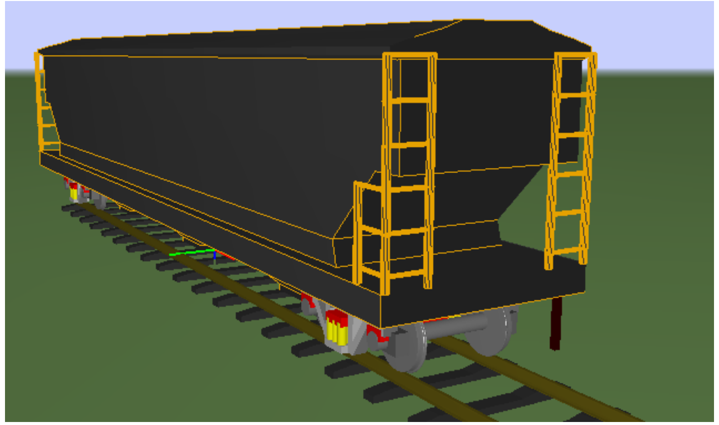
Important wagon inputs:

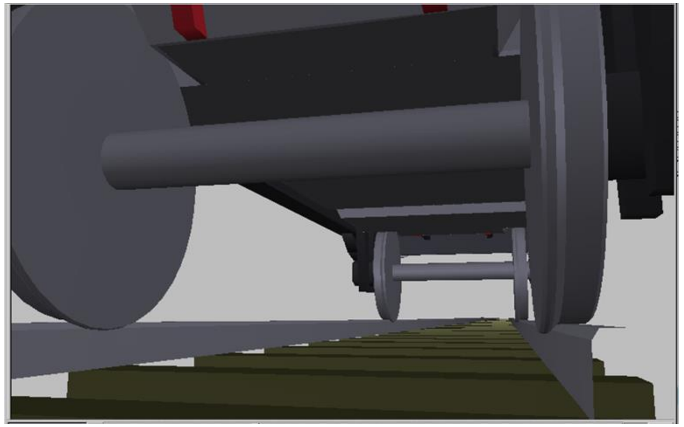
- Car type, length, weight
- Wheel profiles and diameters
- Truck type and spring nest information
- Gib clearances
- Side bearing types and heights

Important track inputs:

- Track geometry
- Rail profiles







Best Practices for Derailment Simulation

1. Simulate Track/Train Dynamics first (even if to rule out if a factor)
2. Apply in-train force to Vehicle/Track Interaction model
3. Replicate derailment (wheel climb, wheel drop, or rail rollover)
4. Then individually adjust parameters back to “normal” conditions to identify root cause.

Thank You to the Technology Venders

QUESTIONS?